Faculty of Engineering

Topics in Computer Science

Bibliometrics and Peer Review

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Sources

Slides taken from Werner Nutt:

- Peer Review

- Bibliometrics

Bibliometrics

- Are there objective (= quantitative) ways to measure
  - the productivity of a researcher?
  - the quality of journals?
  - the productivity of institutions?
- Ideas:
  - Researchers write papers
    - ➔ Analyse the paper output
  - Researchers build upon the work of other researchers, which they cite
    - ➔ Analyse how often papers have been cited

Bibliometric indices

N_p: total number of papers published

N_c: total number of citations received

n_c: mean number of citations per paper (= N_c/N_p)

IF: impact factor, calculated for journals (E. Garfield, 1955)

h-index: Hirsch's h-index (Jorge Hirsch, 2005)

g-index: Response to h-index (Leo Egghe, 2006)

*i*10-index: number of papers with ≥ 10 citations (Google Scholar)

Impact factor of a journal

Published yearly in the *Journal Citation Reports (JCR)*

- by the ISI (= Institute for Scientific Information),
  founded by E. Garfield,
  now part of the *Science and Scholarly Research* division of Thomson-Reuters
- based on two databases with scientific articles, one for science and another one for humanities

Definition of IF_n (= impact factor for year n)

- IF_n = average number of citations in year n
  for articles published in years n-1 and n-2
- IF_n is published in year n+1

Impact factors of some journals in 2013/14

Nature:	42.3/41.5
Science:	31.5/33.6

JACM:	2.939/1.394
ACM TODS	0.750/0.684
VLDB Journal	1.701/1.568
IEEE TOSE	2.292/1.614
TCS	0.516/0.657

JACM	= Journal of the Association of Computing Machinery
ACM TODS	= ACM Transactions on Database Systems
IEEE TOSE	= IEEE Transactions on Software Engineering
TCS	= Theoretical Computer Science

What the IF is supposed to tell us

According to ISI:

- Helps librarians and researchers to decide which journals to subscribe to
- Provides sales arguments for journal publishers
- Supports academic evaluation
  (of researchers, institutions, etc.)

Criticism of IF

- IF can be increased by playing games
  - invite senior researchers
  - reduce types of articles that attract fewer citations
  - publish articles likely to attract citations early in the year

  …
- IFs differ across disciplines
- Articles in a journal with high IF are not necessarily frequently cited
  - Distribution of citations over articles follows a power law
    e.g., 90% of citations in Nature come from 25% of the papers
  - Correlation between paper citation frequency and impact factor is diminishing, since papers are available electronically

    See: Lozano, Larivière, and Gingras. The weakening relationship between the impact factor and papers' citations in the digital age. Journal of the American Society for Information Science and Technology

The h-index

Suggested in 2005 by Jorge E. Hirsch, a physicist at UCSD, as a single index tool for determining a researcher's productivity

Definition: For a multiset (bag) of natural numbers M, we say that

$h(M) := \max \{ n \mid \text{there are } n \text{ elements } c \in M \text{ with } c \geq n \}$

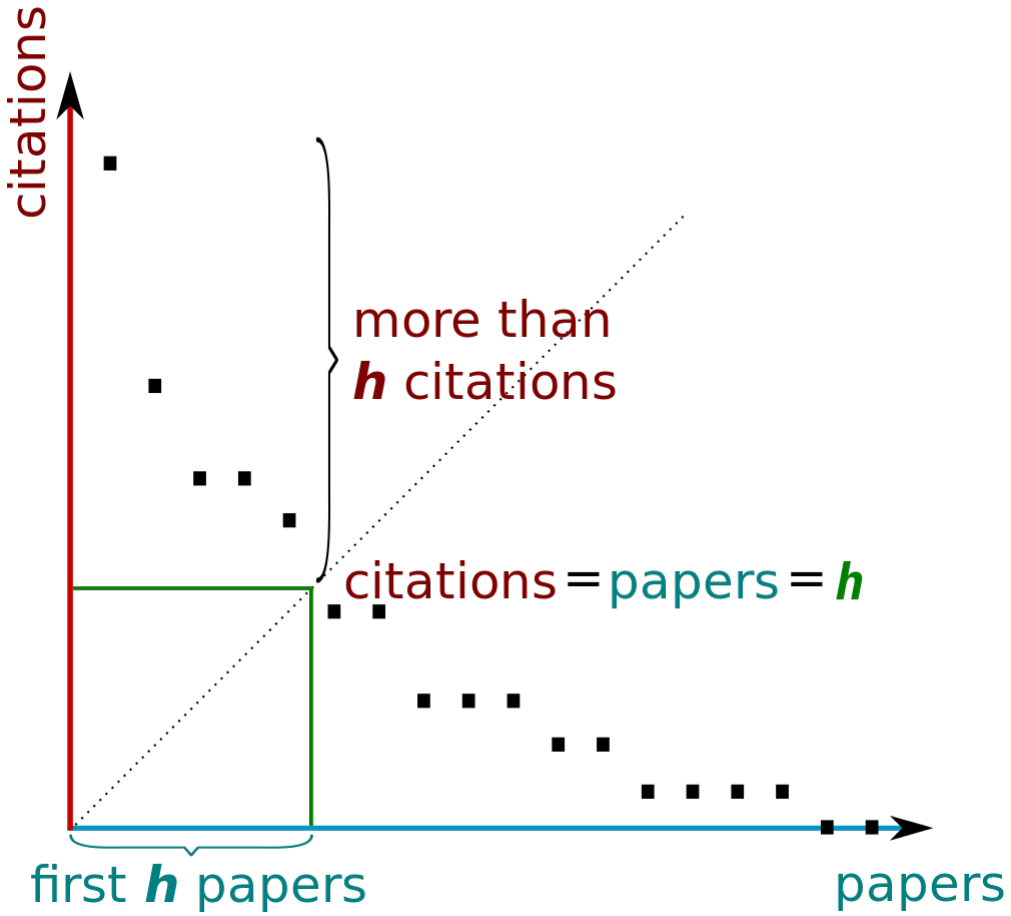
Examples:

$M_1 = \{\{100, 99, 80, 8, 4, 4, 1\}\} \rightarrow h(M_1) = 4$

$M_2 = \{\{5, 5, 5, 5, 4, 4\}\} \rightarrow h(M_2) = 4$

In bibliometrics, the multiset M of an author a is the collection of citation counts of a's papers

The h-index: graphically



Source: Wikipedia

Advantages of the h-index

- It relies on citations to papers themselves, not the journals
- It is not dramatically skewed by a single well-cited, influential paper (unlike total number of citations would be)
- It is not increased by a large number of poorly cited papers (unlike total number of papers would be)
- It minimizes the politics of publication. A high-impact paper counts regardless of where it was published
- It's good for comparing scientists within a field at similar stages in their careers
- It may be used to compare not just individuals, but also departments, programs or any other group of scientists.

Cited from: http://www.benchfly.com/blog/h-index-what-it-is-and-how-to-find-yours/
Blog by Alan Marnett

Criticism of the h-index

The h-index

- favours papers with many authors
- discards information in author placement on author lists
- does not take into account the context of a citation

(e.g., favorable vs. critical,
fleshing out an introduction vs. result or method
enabling work in current paper)

h-index: Summary

"In summary, I have proposed an easily computable index, h, which gives an estimate of the importance, significance, and broad impact of a scientist's cumulative research contributions. I suggest that this index may provide a useful yardstick with which to compare, in an unbiased way, different individuals competing for the same resource when an important evaluation criterion is scientific achievement."

Hirsch, J. E. (15 November 2005). "An index to quantify an individual's scientific research output". PNAS 102 (46): 16569–16572

In reality, there is not “the” h-index

The h-index is always computed over some database:

- Google Scholar
- Scopus (http://scopus.com by Elsevier)
- ISI – Web of Knowledge
  (Microsoft Academic Search)

Databases differ wrt to the publications they cover …

Variants of the h-index:

- over the last x years
- w/ or w/o self-citations
- for institutions

Challenges in creating a citation db: …

How one can dupe Google Scholar …

- Invent a new author, e.g., Marco Alberto Pantani-Contador
- Create 6 faked documents by that author, with lots and lots of citations
- Upload the documents on a university website
- Wait until Google Scholar detects them

Outcome: increase of 774 citations in 129 papers, 6 per cited paper

Google Scholar never corrected the fake citations.

Emilio Delgado Lopez-Cozar, Nicolas Robinson-Garcia, Daniel Torres-Salinas. Manipulating Google Scholar Citations and Google Scholar Metrics: simple, easy and tempting http://arxiv.org/abs/1212.0638

Ranking of Conferences and Journals

CORE (= Computing Research and Education Association of Australasia) is an association of university departments of CS in Australia and New Zealand

- ranks CS conferences
- ranks CS journals

Based on introduction and reranking requests (with detailed arguments) by researchers

- decision about change are also based on Google Scholar and ArnetMiner

See CORE conference ranking
(http://www.core.edu.au/index.php/conference-rankings)

What is a peer?

Merriam Webster

- a member of the British nobility, i.e., a member of one of the five ranks (as duke, marquess, earl, viscount, or baron) of the British peerage

- a person who belongs to the same age group or social group as someone else

What is a peer?

Online etymology dictionary:

- c.1300, "an equal in rank or status" (early 13c. in Anglo-Latin), from Anglo-French peir, Old French per (10c.), from Latin par "equal" (see par (n.)).
- Sense of "a noble" (late 14c.) is from Charlemagne's Twelve Peers in the old romances, who, like the Arthurian knights of the Round Table, originally were so called because all were equal.
- Sociological sense of "one of the same age group or social set" is from 1944.
- Peer review attested by 1970.
- Peer pressure is first recorded 1971.

Peer review in academia

Peers decide about

- which papers are published (editorial peer review)
- which project proposals are funded (funding review)
- *(hiring as a researcher, professor,…*
  the higher the rank, the more peerish the decision)

How does editorial peer review work?

Two models

- Conference Reviewing
- Journal Reviewing

Conference Reviewing

- Program committee (PC)
  - program chairs (appointed by organizing committee of the conference series)
  - PC chairs invite area chairs and other PC members
  - PC members may introduce additional reviewers
- Strict deadlines, short time for reviewing (4-6 weeks)
- Papers are divided among PC
- 5-10 papers per PC member (batch reviewing)
- Decision (in most cases): accept/reject

Journal Reviewing

- Editorial board
  - editor(s) in chief and editorial board members
  - one editor takes care of a submitted paper
  - editor asks ~3 experts for reviews
- Reviewer has 4-6 weeks time for a review (can be extended)
- Decision:
  - accept as is (rare)
  - accept with minor revision (also rare)
  - revision and resubmission
  - reject

Blind vs. Double-blind Reviewing

Blind vs. double-blind reviewing

- reviewers are unknown: *blind* or *single-blind*
- authors are unknown, too: double blind

Double-blind reviewing means: authors must make any reasonable effort to anonymize the paper. i.e.,

- no author names
- no reference to own work

  instead of, “we have shown in [RN11]”,
  say “Razniewski and Nutt have shown in [RN11]”

R. Snodgrass has summarized research on single vs. double-blind reviewing. “Single-Versus Double-Blind Reviewing: An Analysis of the Literature”, SIGMOD Record, 2006

Paper bidding

PC members bid for paper assignment, based on

- title
- abstract
- *(keywords, areas)*
- *(authors and their affiliation)*

PC members express degrees of preference for paper:

- e.g., high, medium, low in EasyChair
- assignment based on preferences and possibly area match

Reviewing proper

PC members

- produce a report
  - write report themselves, or
  - give the task to an auxiliary reviewer (colleague, PhD student, postdoc)
- summarize the report in a (provisional) judgment with grades (e.g., strong accept, accept, weak accept, borderline, weak reject, reject)
- view other reports after having submitted their own
- discuss their reviews and possibly adjust their judgment and review
  - if asked by PC chairs or area chairs (= meta-reviewers) to do so in case the judgments vary (significantly)
- PC/area chairs finally decide about acceptance or rejection

Rebuttal phase

May be inserted between

- first judgment of reviewers and
- final judgment of reviewers

- Authors respond to reviews
  - in short period, e.g., 4 days
  - with limited space, e.g., 500 words
- Reviewers may revise their reviews (and grades) based on other reviews and author rebuttal
- Meta-reviewers report to PC chairs
- PC chairs decide

Revision phase

Recently introduced in some conferences (VLDB, SIGMOD)

(not together with rebuttal)

- Reviewers not only decide about acceptance/rejection, but may also request a revision
- PC chairs/meta-reviewers decide about acceptance, rejection, or revision
- Authors have limited period (approx. 4 weeks) to revise the paper according to comments of reviewers
- Reviewers comment on the outcome of the revision and discuss
- PC chairs/meta-reviewers come up with final decision

Conference Management Systems

- Conference management has typical workflows
- Conference management systems allow organizers
  - to define the specific workflow of for their conference
  - invite a PC
  - receive submissions
  - organize paper bidding and allocation
  - collect of reviews
  - organize discussions
  - let participants of the reviewing process communicate
  - etc.
- Frequently used
  - EasyChair (http://www.easychair.org)
  - Microsoft CMT (https://cmt.research.microsoft.com/cmt/default.html) (see also https://cmt.research.microsoft.com/cmt/userroles.html)

Let's look at examples

- VLDB 2011
  - 3 reviews, acceptance
- KR 2014
  - 3 reviews
  - rebuttal
  - rejection
- SIGMOD 2015
  - 3 reviews
  - revision
  - acceptance

Dynamics of reviewer discussions

- Who is likely to start?
  - The reviewer with the positive review?
  - Or the one with a negative review?

  Hint: Which reviews are usually longer? The positive or the negative ones?
- What is the most probable outcome of a discussion and an adjustment?

Journal Reviewing

In my experience, typically, single-blind

Differences with conference reviewing

- one paper, no batch
- possibility to have a look at the paper before agreeing to review
- possibility to decline a request
- wider range of judgments: accept, minor revision, revision, reject
- interaction with authors: authors explain how they took the reviewers' comments into account for the revision
- iteration: up to 2 revisions possible
- no discussion among reviewers,
  co-reviewers do not reveal their identity

Open reviewing models

- Submissions and reviews are visible to the world,
  reviewers are anonymous
  - e.g., peerj.com (https://peerj.com) (in life sciences and psychology)

    Example paper by Daniel Graziotin et al.
    (https://peerj.com/articles/289/)

- Submissions and reviews are visible to the world,
  reviewers identify themselves (but may remain anonymous),
  other researchers may contribute reviews,
  all contributing reviewers are acknowledged in the end
  - e.g., Semantic Web Journal

Structure of a review

- Summary of the paper
- Character of the work, relevance for outlet
- Contribution of the paper, relationship to state of the art
  - originality
  - depth
- Strengths
- Weaknesses
- Writing
- Explanation of judgment
- Detailed comments

See also Allen S. Lee "Reviewing a Manuscript for Publication" (http://www.people.vcu.edu/~aslee/referee.htm)

Review Structure CIKM, DB Track

- Overall Rating
  - Strong Accept, Accept, Neutral, Reject, Strong Reject
- Top 3 Strengths
- Top 3 Weaknesses
- Detailed Comments
- Author feedback needed?
- What specific feedback do you like the authors to provide
- Comments to Program Chair

Review Structure IJCAI

- Summary: Describe the paper in 2-3 sentences
- Relevance: Is the work relevant to AI?
- Originality: Does the paper clearly point out differences from related research? Are the problems or approaches new?
- Significance: Is the work important? Does the paper make a valuable contribution to knowledge and understanding in the AI area? Does it advance the state of the art? Does the paper add to our understanding of some aspect of agent systems? Does the paper stimulate discussion of important issues or alternative points of view? Does the paper carefully evaluate the strengths and limitations of its contributions, and draw lessons for future work?
- Technical Quality: Is there a careful evaluation of the proposed method and the results? Is the paper technically sound, with compelling arguments?
- Readability and Organization: Is the paper clearly written? Does the paper motivate the research? Are results clearly described and evaluated? Is the paper well organized?

Review Structure VLDB

- Overall Recommendation: Accept, Revise, Reject
- Summary: what is proposed? in what context? brief justification of recommendation
- 3 strong points
- 3 weak points
- Relevance
- Novelty
- Significance
- Technical Depth, quality of content
- Presentation
- Would you champion acceptance?
- Detailed comments

Peer Review for Research Funding

Funding bodies/agencies base decisions on peer reviews, e.g.,

- South Tyrol:
  - internal projects of UNIBZ, evaluated by Univ. Research Committee 1mio/yr
  - calls by province for research and innovation projects ~2-3 mio/yr
- Italy:
  - PRIN ~100 mio/yr, FIRB (no funding for both in the last few years)
- Austria:
  - FWF (= Fonds zur Förderung der wissenschaftlichen Forschung) 180 mio/yr
  - also FFG, AWS, …
- Germany:
  - DFG (= Deutsche Forschungsgemeinschaft) 2.6 billion EUR/year
    22,000 reviews/yr
  - BMBF (= Bundesministerium für Bildung und Forschung)

Peer Review for Research Funding/2

Funding bodies/agencies base decisions on peer reviews, e.g.,

- UK: Research Councils
  - EPSRC (= Engineering and Physical Sciences Res. C.) 800 mio GBP/yr 2000 proposals/yr
  - BBSRC (= Biotechnology and Biological Sciences Res. C.) 450 mio GBP/yr
  - all councils: combined budget 3.5 bio GBP/yr, out which 1 bio for grants
- France:
  - ANR (Agence Nationale de la Recherche) 1 billion EUR
- Belgium:
  - FSO (= Fonds Wetenschappelijk Onderzoek) in Flanders, 200 mio EUR/yr
  - FNRS (= Fonds de la Recherche Scientifique) in Wallonie
- USA:
  - NSF, DARPA, NIH, …
- EU:

Structure of a Grant Proposal

- Problem to be solved, background, related work
- Significance
- Research outcomes, e.g., theorems, prototypes, emp. study
- Impact on research community, economy, society
- Approach, work programme
  - Work packages: tasks, deliverable, milestones responsibilities, how measure success
- Resources: how much money for what?
- Investigators (PI, coI) team: competence, track record
- Management: coordination, risk management

Workflow

- Applicant submits project proposal
- Agency nominates reviewers
  - proposed by staff
  - approved by academic on boards
- Panel of researchers ranks proposals
  - 1 rapporteur for proposal reports on reviews
  - top ranked proposals are funded

CONCLUSION

review?

- Discussion

decision about acceptance/rejection?

- Discussion